

Zhenyu Wu

Applied Scientist — LLMs, Structured Data & Knowledge Graphs

 Manchester, UK
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Summary

Final-year Computer Science PhD candidate specialising in LLM-based structured-data understanding, schema and taxonomy inference, knowledge graphs, and agentic data analytics. Experienced in building reproducible research systems with PyTorch, Hugging Face Transformers, vLLM, and LangGraph.

Experience

University of Manchester , Doctoral Researcher — LLMs for Structured Data and Knowledge Construction	Manchester, UK Sept 2022 – Oct 2026 4 years 2 months
- Designed encoder- and decoder-based systems for inferring entity types, attributes, hierarchies, and semantic relationships from heterogeneous tabular repositories. - Improved F1 and Recall by 30–40% and Rand Index by 7–10% over prior approaches across real-world datasets. - Built modular Python pipelines with PyTorch, Hugging Face Transformers, and vLLM for batched inference, caching, structured validation, ablations, and reproducible evaluation. - Developing an agentic multi-database question-answering system combining schema linking, structured reasoning, Text-to-SQL, and tool calling with Qwen2.5-14B.	
Singapore University of Technology and Design , Visiting Research Associate	Singapore Apr 2021 – Mar 2022 1 year
- Co-developed a unified benchmark for data-stream clustering, published in Proceedings of the ACM on Management of Data (SIGMOD 2023). - Implemented CluStream, DenStream, DStream, and DBStream within a shared C++ code-base and reproducible evaluation framework.	

Education

PhD University of Manchester , Computer Science	Manchester, UK Sept 2022 – Oct 2026
- Expected completion October 2026 - Dean's Doctoral Scholarship	
MSc University of Edinburgh , High Performance Computing with Data Science	Edinburgh, UK Sept 2019 – Aug 2020
Graduated with Merit	
BSc Tianjin University of Finance and Economics , Information Management and Information Systems	Tianjin, China Sept 2015 – June 2019
- GPA: 3.81/4.00; Top 5% - Minor in Accounting	

Publications

Schema Inference for Tabular Data Repositories Using Large Language Models Zhenyu Wu , Jiaoyan Chen, Norman W. Paton arxiv.org/abs/2509.04632 (The VLDB Journal)	Jan 2026
Taxonomy Inference for Tabular Data Using Large Language Models Zhenyu Wu , Jiaoyan Chen, Norman W. Paton 10.1007/978-3-031-94575-5_22 (The Semantic Web — ESWC 2025)	Jan 2025

Dataset Discovery and Exploration: A Survey

Jan 2024

Norman W. Paton, Jiaoyan Chen, **Zhenyu Wu**[10.1145/3626521](#) (ACM Computing Surveys)**Data Stream Clustering: An In-depth Empirical Study**

Jan 2023

Xin Wang, Zhengru Wang, **Zhenyu Wu**, Shuhao Zhang, Xuanhua Shi, Li Lu[10.1145/3589307](#) (Proceedings of the ACM on Management of Data)

Skills

Programming: Python, SQL, C++, Linux/Shell, Git**Machine Learning and LLMs:** PyTorch, Hugging Face Transformers, vLLM, self-supervised learning, contrastive learning, hierarchical clustering**Structured Data and Knowledge:** Schema and taxonomy inference, ontology construction, knowledge graphs, table understanding, entity typing, attribute resolution**Agents and Data Intelligence:** LangChain, LangGraph, tool calling, structured-output validation, schema linking, Text-to-SQL

Awards

Dean's Doctoral Scholarship

University of Manchester

Jan 2022

Languages

Mandarin: Native**English:** Professional proficiency